

PROPAGATING PARTICLES MODEL OF UNCONSTRAINT EXAMINATION TIMETABLING

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Abstract

Construction of feasible examination timetables implies strong interaction between the involved exams. We propose a dynamical model of unconstraint examination timetabling, where exams are represented by interacting particles cyclically moving over the timeslots. A simple interaction rule is adopted – move to the next slot, if in a clash, and the corresponding evolutionary operators are derived. The model is applied to the Toronto benchmark problems and evolution of the resulting systems is studied in terms of the number of clashes. Based on the observed main properties, a Propagating Particles Algorithm of an Academic Time Tabling (bi-PAT) is formulated without explicitly drawing on search or optimization paradigms. The algorithm is compared with 3 other methods originating from analogies with natural science phenomena, as well as 5 combinatorial algorithms. Competitive performance of bi-PAT with the best algorithms is observed in case of problems of moderate size. The reasons of significant deviation from the best results in case of large problems are discussed and a brief outline of the future developments is given.

Keywords: examination, timetabling, model, force, dynamics, unconstraint

1. Introduction

Timetabling problems generally and educational timetabling specifically have been subjects of active studies for many decades due to their high practical value. Different formulations of

educational timetabling problems exist. In the survey of automated timetabling [1] they are classified into three main types – school, course and examination timetabling. It is added, however, that such division is not strict, and hybrid cases can appear in practice. Differences between the mentioned types are discussed in [2]. In the present paper we consider an unconstrained examination timetabling problem with the only objective to avoid students taking different exams at the same time. Usually, additional hard and soft constraints are imposed to such time tables. Several important examples are enlisted in [3].

Many methods have been proposed since 1960s. The algorithms and research directions are summarized in an in-depth survey of examination timetabling methodologies [3]. Ordering heuristic approaches originate from the fundamental work [4], in which timetabling is treated from a graph colouring point of view. Such methods sort the exams based on respective criteria and sequentially construct corresponding timetables. Five sorting strategies are discussed and compared in [5]: by saturation degree – the number of available timeslots [6]; by largest degree – the number of conflicts; by weighted largest degree – the number of students in conflict; by largest enrolment; and, finally, by random ordering. It is shown that, generally, sorting by saturation degree combined with backtracking strategy leads to more compact schedules and requires less computing time.

Combinatorial algorithms developed by analogies with natural science phenomena prove to be reasonably productive. Simulated annealing is one of local search methods widely used in scheduling and timetabling. It originates from an equation of state calculation algorithm for systems composed of pair wise interacting molecules with spherically symmetric potential filed [7]. A hybrid method combining simulated annealing and constraint programming is

proposed in [8]. Being tested on the same data the approach results in somewhat longer unconstrained timetables than those constructed in [5].

Genetic algorithms mimic natural evolutionary processes and the principles of genetics. Possible solutions are represented as chromosomes, and new chromosomes are generated by crossover and mutation mechanisms. Applications of genetic algorithms to educational timetabling problems are summarized in a brief survey [9]. An algorithm with linear linkage encoding representation scheme is suggested in [10] for solving the graph colouring and exam timetabling problems. Another class of methods inspired by a biological concept consists of ant colony algorithms that behave similar to ants in their search for the shortest path to food. Such an approach is developed in [11] and used as an evidence that ant colony optimization should not be limited to routing problems, though being firmly focused to the latter. It is shown that proper configuration puts the algorithm among the best methods applied to the same unconstrained problems as in [5].

Suggesting potential field or force directed algorithms for solving combinatorial problems is not a new idea. Essentially topological areas of application benefit most from such methods. By far incomplete list of examples includes graph drawing [12], placement for 3D VLSI [13], routing in wireless ad hoc networks [14] and robotics [15], and many more. Graph drawing is closely related to task scheduling. It is a common and useful practice to represent the latter as a task graph with the tasks being vertices connected by edges corresponding to inter-task dependencies. Force-directed clustering heuristics is implemented in [16] by substituting tasks with charged particles and the edges in the task graph – with springs. It is shown that the force-directed approach outperforms other clustering heuristics, especially for a limited number of task processors.

Timetabling implies strong interaction between the exams through clashes. The present work aims at modelling of this interaction and construction of the corresponding dynamical system. The main goal is to show that adequate dynamical description can lead to feasible solutions without explicitly drawing on search or optimization heuristics. If succeeded, such approach will allow us to utilise the well developed formalism of classical mechanics [17] in timetabling problems and reveal certain characteristics of solutions before entering computationally expensive experimental and testing phases.

2. Dynamical Model of Examination Timetabling

Let us consider construction of a feasible examination time table – given N class lists, schedule the corresponding N exams for S timeslots in a clash free way. Two exams clash, if they are scheduled for the same slot and there is at least one student included in both class lists. Let us then consider the exams as a 1D dynamic system with the enumerated timeslots being the discrete periodic coordinate of period S , and adopt a simple interaction rule – if in a clashing state, an exam shifts to the next slot. Formally, we introduce a force operator $\mathbf{F}_k$ that acts on the k -th exam and has the following form:

$$\mathbf{F}_k(t_k) = (t_k + b) \bmod S \quad (1)$$

where t_k is the current timeslot of the k -th exam ($0 \leq t_k < S$ and $0 \leq k < N$); and binary constant b is equal to 1, if the exam is in a clashing state at t_k , and 0 – otherwise.

Generally, operators $\mathbf{F}_k$ do not commute and, therefore, exam numbering matters:

$$\mathbf{F}_k(t_k) \otimes \mathbf{F}_n(t_n) \neq \mathbf{F}_n(t_n) \otimes \mathbf{F}_k(t_k) \quad (2)$$

Indeed, if we imagine just three exams enumerated 0, 1, 2 and scheduled for the same timeslot, such that 0 clashes with 1, and 1 with 0 and 2, then $\mathbf{F}_0(0) \otimes \mathbf{F}_1(0) = 1 \otimes 1$, while $\mathbf{F}_1(0) \otimes \mathbf{F}_0(0) = 1 \otimes 0$.

In order to generalize the rule of motion (1) and allow moves of arbitrary depth, we also consider integer powers of $\mathbf{F}_k$:

$$\mathbf{F}_k^m(t_k) \equiv \mathbf{F}_k \otimes \mathbf{F}_k \otimes \dots \otimes \mathbf{F}_k \text{ } m \text{ times} = (t_k + c) \bmod S \quad (0 \leq c \leq m) \quad (3)$$

where c is the number of successive timeslots starting from t_k , for which exams clashing with the k -th one are scheduled. If, for example, all slots are occupied by such exams, then $\mathbf{F}_k^{S-1}$ will simply shift the latter one slot backward:

$$\mathbf{F}_k^{S-1}(t_k) = (t_k - 1) \bmod S \quad (4)$$

Successive action of these operators will determine the evolution of the entire system we will be tracking down in terms of changes in the number of clashes. Therefore, we introduce parameterized cumulative evolutionary operators $\mathbf{E}$ of the following form:

$$\mathbf{E}(m, d) = \left(\prod_{k=0}^{N-1} \mathbf{F}_k^m(t_k) \right)^d \quad (5)$$

where m is called mode and $d \gg 1$ – duration. In expanded terms, $\mathbf{E}(m, d)$ shifts in succession each of N exams up to m modulo S slots forward subject to clashes and repeats this cycle d times.

Fig. 1 illustrates the action of $\mathbf{E}(m, d)$ on real life timetable examples taken from the repository widely known as Toronto dataset, which was first introduced in [5] and later refined and standardized in [3]. The overall number of clashes left in the system after each of d iterations is plotted against the mode duration d . In all cases exams are initially scheduled for the first timeslot. Obviously, this is the state with the maximal number of clashes.

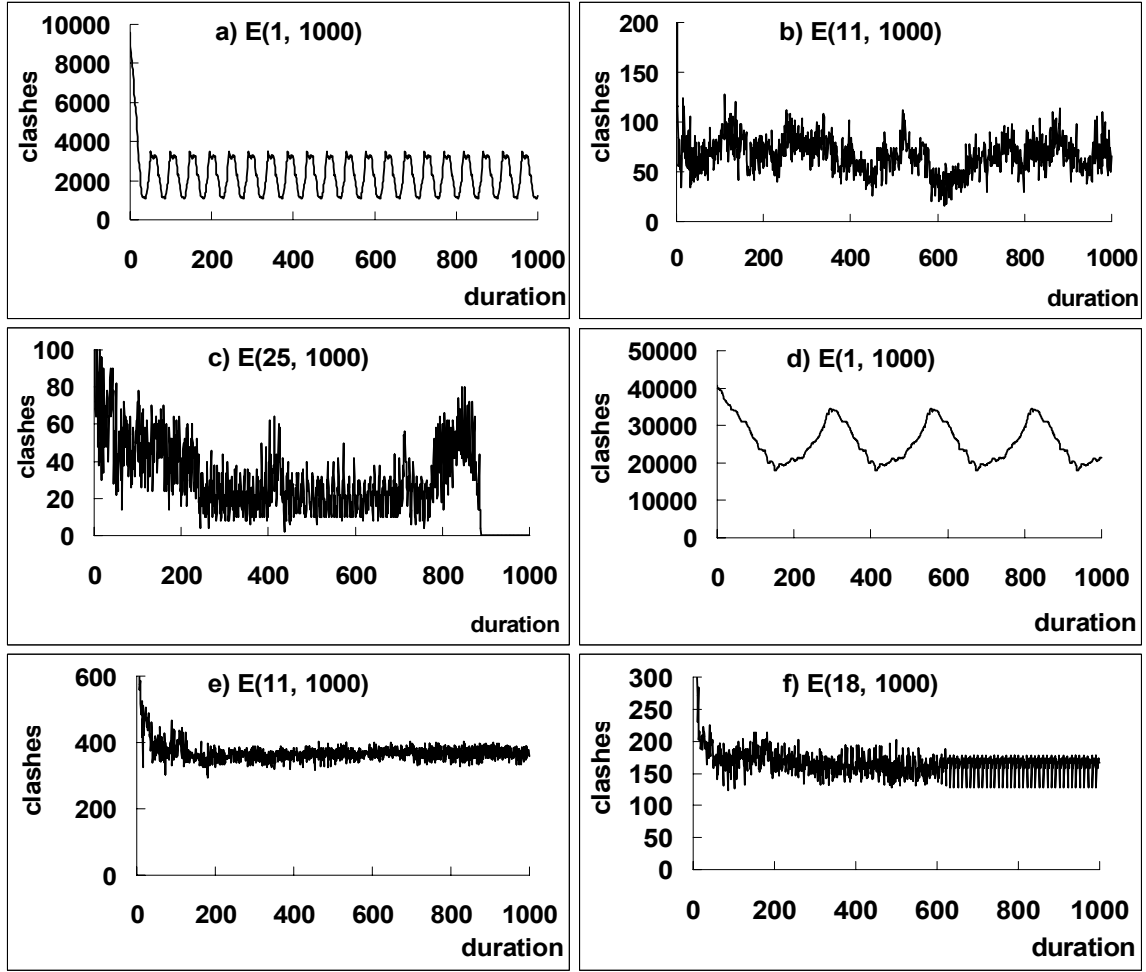


Figure 1. Examples of $E(m, d)$ cumulative operators: a), b) and c) acting on a small timetable ear-f-83; d), e) and f) – on a large timetable car-f-92.

Fig. 1a, 1b and 1c correspond to “ear-f-83.stu” data file – a relatively small timetable with 190 exams, 24 slots and 9586 clashes in the initial state, while Fig. 1d, 1e and 1f – “car-f-92.stu” with 543 exams, 32 slots and 40610 clashes.

Dynamics of such systems passes through an initial transient stage followed by steady-state periodic phase. Therefore, for given duration d , we can distinguish between three mode types: periodic modes dominated by steady-state phase $d_{\text{transient}} \ll d_{\text{steady-state}}$ (Fig. 1a and 1d); quasi-periodic modes effectively limited to the transient phase $d_{\text{transient}} \approx d$ (Fig. 1b, 1c and

1e); and mixed modes with comparable stages $d_{\text{transient}} \approx d_{\text{steady-state}}$ (600 vs. 400 iterations in Fig. 1f).

Any superposition of $\mathbf{E}(m, d)$ operators that constructs a clash free examination timetable will identify a timetabling model. Particularly, the model can be reduced to a single operator.

3. Propagating Particles Algorithm of an Academic Time Tabling bi-PAT

In the current work we create timetabling models based on the Toronto benchmark dataset. Its problems are summarized in Table 1 sorted by the maximal number of clashes – the main measure of complexity we adopt in the current paper.

1. Toronto Benchmark Dataset

#	Problem	Exams	Students	Clash density	Max clashes	Timeslots
1	hec-s-92	81	2823	20%	2726	18
2	sta-f-83	139	611	14%	2762	13
3	ute-s-92	184	2749	8%	2860	10
4	lse-f-91	381	2726	6%	9062	18
5	yor-f-83	181	941	27%	9412	21
6	ear-f-83	190	1125	29%	9586	24
7	kfu-s-93	461	5349	6%	11786	20
8	tre-s-92	261	4360	18%	12262	23
9	rye-s-93	486	11483	7%	17744	23
10	car-f-92	543	18419	14%	40610	32
11	uta-s-93	622	21266	13%	48498	35
12	car-s-91	682	16925	13%	59628	35
13	pur-s-93	2419	30029	3%	164598	42

In order to get insight to the basic properties of $\mathbf{E}(m, d)$, we schedule all the exams of the given problem for the first timeslot – the initial state, and separately apply $\mathbf{E}(m, 1000)$ for each of $1 \leq m < S$. Fig. 2 depicts the minimal number of clashes as a function of m for 8 relatively small problems from Table 1. Data points corresponding to periodic and mixed modes are in white, to quasi-periodic ones – in black.

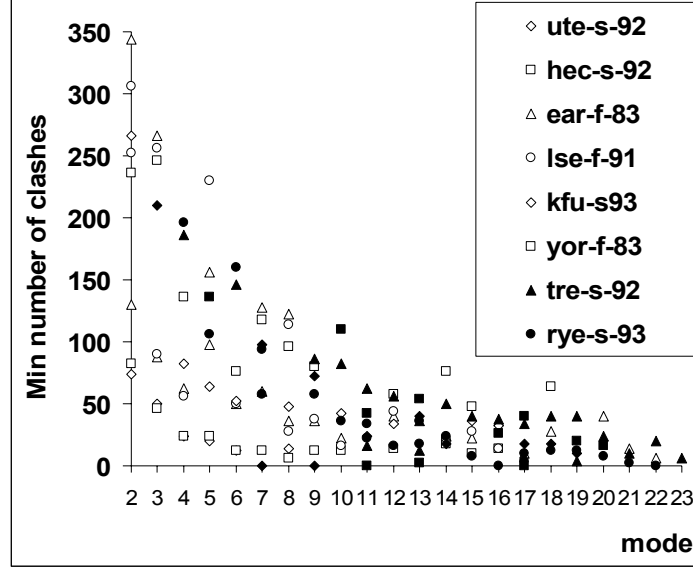


Figure 2. Dependence of the minimal number of clashes on modes ($d = 1000$) in small Toronto benchmark problems.

There can be drawn certain observations connecting the values of m with the mode type. If m and S are co-prime numbers, then the corresponding modes are quasi-periodic. The only exception is $m = 2$, where the steady-state periodic phase is always present, at least for duration $d = 1000$, even if S is odd. All other modes are in resonance with S and, therefore, are either periodic or mixed. Particularly, if m is a factor of S , then corresponding modes are strictly periodic.

Secondly, there may be quasi-periodic modes that construct feasible timetables of the given problem. We call them feasible modes. For example, both quasi-periodic modes of ute-s-92 problem $m = 7$ and $m = 9$ are feasible (note that $m = 3$ is a mixed mode). Obviously, duration of the feasible modes is shorter for simpler problems. Table 2 summarizes feasible modes not longer than $d = 100000$ for the first 9 problems of Table 1. These are the smallest timetables with less than 25 slots and the maximal number of clashes not exceeding 20000.

There is one special case – sta-f-83: all its modes are feasible with the duration decreasing from $E(1, 12)$ to $E(12, 1)$.

Table 2. Feasible Modes of the Small Toronto Benchmark Problems

Problem	Timeslots	Feasible Modes			
hec-s-92	18	$E(11, 96)$	$E(17, 100)$		
sta-f-83	13	All modes, $1 \leq m \leq 14, d \leq 12$			
ute-s-92	10	$E(7, 22)$	$E(9, 30)$	$E(11, 90)$	
lse-f-91	18	$E(17, 40010)$			
yor-f-83	21	$E(19, 9223)$	$E(20, 13502)$		
ear-f-83	24	$E(11, 74002)$	$E(13, 3615)$	$E(17, 3106)$	$E(19, 2915)$
		$E(23, 2606)$	$E(25, 889)$		
kfu-s-93	20	$E(17, 4658)$	$E(19, 58134)$	$E(21, 15774)$	
tre-s-92	23	$E(17, 85639)$	$E(18, 88369)$	$E(19, 10570)$	$E(20, 22989)$
		$E(21, 42432)$	$E(22, 77780)$	$E(24, 19729)$	
rye-s-93	23	$E(12, 65518)$	$E(13, 8585)$	$E(14, 5066)$	$E(15, 10274)$
		$E(16, 836)$	$E(17, 1732)$	$E(18, 1799)$	$E(19, 9705)$
		$E(20, 3591)$	$E(21, 1007)$	$E(22, 797)$	$E(24, 10512)$

Assuming that there is a solution for the given number of timeslots S , the obtained results allow formulating a modal version of a Propagating Particle Algorithm of an Academic Time Tabling – bi-PAT, as follows:

1. Construct the initial schedule by assigning all exams to the first timeslot.
2. Select a sufficiently large value of d and apply different $E(m, d)$ operators to the initial schedule, where m and S are mutually prime numbers and $2 < m < 2S$.
3. If none of the applied operators constructs a feasible timetable, double d and repeat the step 2.

Implementation of bi-PAT in this way may require significantly large duration, since no upper bound for d is explicitly specified. For example, $d = 40010$ for lse-f-91 problem; and $d = 1000000$ is not sufficient for construction of a feasible ear-f-92 timetable – the first large problem in Table 1. It is, therefore, important to consider superposition of different modes. Let us introduce two modal characteristics – the mean number of clashes C_0 averaged over

the entire duration and the amplitude of periodic oscillations or quasi-periodic fluctuations A .

Numerical experiments revealed the following main properties:

- Expectedly, C_0 practically does not depend on d and is a decreasing function of m , $1 \leq m < S$.
- Generally, periodic modes oscillate around lower mean than quasi-periodic ones: $C_{0 \text{ periodic}} < C_{0 \text{ quasi-periodic}}$. Amplitudes of periodic oscillations, however, are smaller than the mean by at least an order of magnitude: $A_{\text{periodic}} \ll C_{0 \text{ periodic}}$. Therefore, in periodic modes the minimal number of clashes $C_{0 \text{ periodic}} - A_{\text{periodic}}$ does not vanish, unless the mean is of order of unity.
- Unlike periodic modes, amplitudes of quasi-periodic fluctuations depend on duration, and for sufficiently large d can become comparable with the mean, thus considerably minimizing the number of clashes (Fig. 1b) or completely resolving them (Fig. 1c).

Aiming at bringing together all positive tendencies of the enlisted properties, we come up with an alternative formulation of bi-PAT – its cumulative version:

1. Select a value of d , large enough to enter steady-state phase of periodic modes;
2. Construct the initial schedule by assigning all exams to the first timeslot, and apply $\mathbf{E}(1, d)$;
3. In a loop for each of $m < S$ modes apply $\mathbf{E}(m, d)$ to the schedule that minimized clashes during the previous iteration $\mathbf{E}(m-1, d)$.

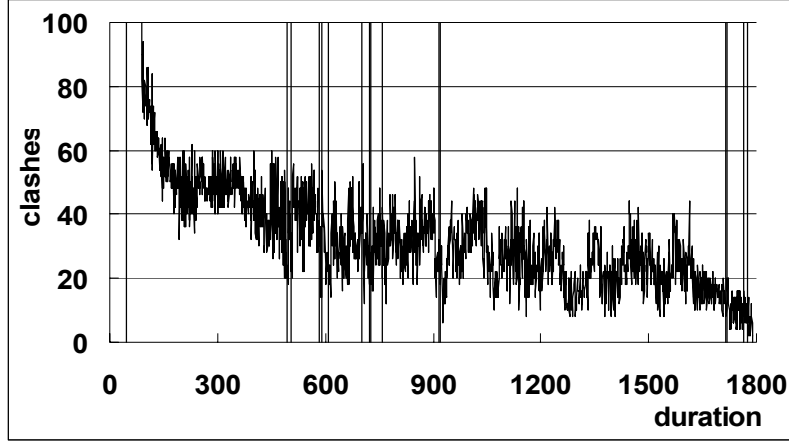


Figure 3. Cumulative bi-PAT solution of the hec-s-92 timetable. The vertical lines indicate the mode bounds.

As an example, the solution of hec-s-92 problem is depicted in Fig. 3. All $E(m, d)$ operators acted for $d = 1000$. For each of them the state with minimal clashes was chosen as the initial schedule for the next one. Vertical bounds indicate the resulting mode duration. Fig. 3 shows that the superposition of the modes adds steadily decreasing behaviour to system's evolution.

Normally, the overall duration of the cumulative version is much longer than the shortest feasible mode. In some cases, however, reversed situation is observed. Particularly, for lse-f-91 data file the cumulative bi-PAT converges in just 3105 iterations compared to 40010 iterations of the only feasible mode $m = 17$ in the range $d \leq 100000$.

4. Testing Results and Discussion

One of the main objectives of the Toronto dataset is to minimize the number of timeslots needed for construction of unconstrained timetables [3]. We compare bi-PAT with the best results achieved by several other methods in Table 3. The five sequential algorithms with backtracking [5] are denoted as follows: LD – sorting by the largest degree, SD – by the saturation degree, WD – by the weighted largest degree, LE – by the largest enrolment, and

RO – random ordering. The simulated annealing hybrid method [8] denoted as SA, the genetic algorithm [10] – GA, and the ant colony optimization method [11] – AC, are also included. The records are divided into two main parts – the first 9 small problems with less than 20000 clashes each, and the remaining 4 large problems.

Table 3. Minimal number of timeslots obtained by different approaches

Problem	LD	SD	WD	LE	RO	SA	GA	AC	bi-PAT
hec-s-92	18	17	17	17	17	18	17	17	17
sta-f-83	13	13	13	13	13	13	13	13	13
ute-s-92	10	10	10	10	10	11	10	10	10
lse-f-91	17	17	17	17	17	18	17	17	17
yor-f-83	20	20	20	19	20	23	20	19	21
ear-f-83	22	22	22	22	22	24	23	22	23
kfu-s-93	19	19	19	20	19	21	20	19	20
tre-s-92	22	20	20	22	22	21	21	20	23
rye-s-93	21	21	22	22	22	22	23	21	22
car-f-92	31	28	30	31	32	31	36	28	35
uta-s-93	33	32	33	33	34	32	38	30	38
car-s-91	32	28	30	31	32	30	36	28	38
pur-s-93	36	35	38	38	N/A	N/A	N/A	N/A	48

All the small timetables have been constructed for timeslots enlisted in Table 1 by both modal and cumulative versions of bi-PAT. However, there are only two cases among them that correspond to the best solutions – sta-f-83 and ute-s-92 with 13 and 10 timeslots respectively. In order to construct timetables with less slots, we apply customised versions of bi-PAT, where different combinations of the $E(m, d)$ operators are empirically selected and tried instead of the cumulative superposition. Here again, the state with minimal number of clashes of the previous mode becomes the initial one for the next evolutionary operator. Particularly, cyclic application of $E(15, 1000)$, $E(16, 1000)$, $E(18, 1000)$ and $E(19, 1000)$ operators constructs hec-s-92 timetable of 17 slots. Another customised bi-PAT improves rye-s-93 timetable: $E(21, 2000) \otimes E(11, 10)^3 \otimes E(21, 100)$. Consideration of larger duration may be an alternative strategy. For example, ear-f-83 timetable of 24 slots is constructed by $E(25, 889)$, while the one with 23 timeslots – by $E(25, 1227)$. Similarly, lse-f-91 timetable of

18 slots is constructed by cumulative bi-PAT with $d = 1000$, while the one with 17 timeslots requires 10 time larger duration $d = 10000$.

Before starting the comparative analysis, let us note that the best results on the Toronto data are achieved by SA [5] and AC [11]. It follows from Table 3 that bi-PAT demonstrates reasonable performance on the relatively small problems. It matches the best results reported for the 4 smallest problems [5, 10-11]. As for the remaining 5 small problems, it is slightly worse than the sequential algorithms [5] and the ant colony method [11], but competitive with SA [8] and GA [10]. We find it important to emphasize that all the mentioned approaches are explicitly based on search or optimization paradigms, while the versions of bi-PAT reflect on evolution of the dynamical models (5).

Different behaviour of the dynamical models is observed in case of the 4 large timetables with more than 40000 clashes. Only cumulative bi-PAT is applied to these problems, since the duration of feasible modes (if any) quickly grows with the problem complexity. The results obtained for car-f-92, uta-s-93 and car-s-91 timetables agree with those achieved by GA [10]. Both methods, however, are significantly outperformed by other approaches [5, 8, 11].

The largest timetable pur-s-93 is a special case. It requires 48 timeslots to construct a feasible timetable by cumulative bi-PAT with $d = 1000$, which is essentially inferior to the sequential algorithms excluding the one of random ordering [5]. At the same time, it is wrong to claim that further improvement is impossible, before studying customized versions of bi-PAT and larger values of duration d . Let us note that no results have been reported by the SA, GA and AC heuristics, as well as the randomly ordering sequential algorithm with

backtracking RO. Thus, in the conducted experiments we prioritized the construction of this timetable over its optimality.

5. Conclusions

In the present work we treat examination timetabling as a system of interacting particles and suggest an iterative and spatially periodic multimodal model of the interaction between the exams through the parameterised evolutionary operators $\mathbf{E}(m, d)$ (1) – (5). Originating from an intuitive concept of moving away from clashing counterparts, the model exhibits realistic behaviour with certain similarities to dynamical systems governed by laws of classical mechanics. Particularly, we observe such well-known and studied phenomenon as resonant orbits or periodic modes, where the period of motion of individual particles m has common divisors with the problem’s spatial period S – the number of available timeslots. Meantime, experiments with the Toronto benchmark problems demonstrate that quasi-periodic modes with co-prime m and S are closely related to the construction of feasible timetables. Each of 9 relatively small problems has at least one quasi-periodic mode of evolution that resolves all the clashes. Based on the revealed general properties, the Propagating Particles Algorithm of an Academic Time Tabling – bi-PAT, is formulated in two versions. In terms of constructing timetables of minimal length, it can compete with other approaches, when applied to problems of moderate size.

It is shown that superposition of $\mathbf{E}(m, d)$ operators different from the modal and cumulative formulations of bi-PAT can enhance the latter’s performance. We managed to find such solutions for 4 timetables. A successful empirical search, however, cannot be practically conducted in large timetables. Therefore, encouraged by the reported initial results, in the next stage of our research we will start developing a self-consistent theory of

the dynamical model of examination timetabling with the ultimate goal to create adequate tools that will describe the solutions at sufficiently abstract level before entering the demanding computational phase.

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